

Exact Three-Target First-Passage Coupling

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Abstract

We compute the exact joint first-passage-time law for every unordered triple of twenty frozen targets on uniform permutations of sixteen labels. A nested-chain closure dynamic program and three-dimensional finite differences certify 1,140 arrays with 17^3 cells, together 5,600,820 permutation-count cells. Pair and single marginals, exact moments through order three, covariances, and third interaction cumulants are independently checked. The result is finite combinatorics under `NO_BAD_EULER_OR_ROOT_NUMBER`.

1 Joint survival and inversion

Let $T_i \in \{0, \dots, 16\}$ and

$$S_{ijk}(a, b, c) = \#\{\pi : T_i > a, T_j > b, T_k > c\}.$$

After sorting thresholds, the relevant supports form a nested chain $A \subseteq B \subseteq C$. Closure states are propagated label by label and contracted with the target inclusion matrix. Boundary planes are supplied by the C88 single-target and C90 pair receipts. Three-dimensional Mobius inversion gives

$$N_{ijk}(a, b, c) = \Delta_a \Delta_b \Delta_c S_{ijk}(a, b, c).$$

Theorem 1. *For all 1,140 unordered target triples, N_{ijk} is nonnegative and sums to $16!$. Every pair and single marginal agrees with the frozen lower-order receipt, coordinate permutations agree, and all raw moments with exponents at most three are exact. The covariance matrix and third interaction cumulant computed from those moments are exact rational values.*

Proof. The closure dynamic program partitions the 4^{16} nested-support chains, and factorial weights count all permutations with each chain of support sizes. The boundary substitutions are the defining C88/C90 survival counts. Finite Mobius inversion is therefore the exact pushforward PMF. Marginalization and coordinate permutation are identities of the same finite count; moments and cumulants are finite sums of integer counts divided by $16!$. $\square$

certified object	count
unordered target triples	1,140
joint PMF cells	5,600,820
pair / single marginal recoveries	400 / 20
mixed-moment cells	72,960
covariance and interaction records	1,140

2 Scope

This package is a finite probability certificate. It makes no arithmetic or local-data, Euler-factor, root-number, automorphy, full Burnside/table-of- marks, or Hilbert–Pólya operator claim.